Accurate Modeling of Railway Turnout Systems and High-Precision State Estimation Methods

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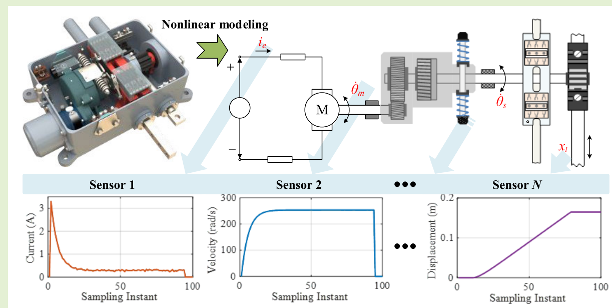
***Abstract*—The railway turnout system (RTS) is a critical component of modern, complex railway transportation networks, where its reliability and safety are essential for ensuring operational efficiency and transportation safety. To address the issue of inaccurate modeling in current turnout systems, this article establishes a precise RTS model based on bond graph (BG) theory that incorporates the sensor model, while considering the effects of gear backlash and nonlinear friction. A nonlinear state-space model of the RTS is further developed by integrating it with a constructed measurement model. Additionally, this article introduces an improved extended Kalman filter (EKF) design method for high-precision state estimation of the nonlinear model. This filter incorporates second- and third-order statistical information derived from nonlinear partial Taylor expansions, significantly enhancing estimation accuracy. In the experimental section, the impact of nonlinear factors on model accuracy is analyzed by comparing data from the model's detectors with actual sensor data, utilizing ablation experiments. Furthermore, the superior performance of the proposed method in nonlinear filtering is validated through the estimation results of primary state variables in the turnout system.**



***Index Terms*— Bond graph (BG), high-precision state estimation, nonlinear filtering design, nonlinear modeling, railway turnout system (RTS).**

I. Introduction

IN THE increasingly complex railway transportation network, the railway turnout system (RTS) is critical infrastructure for multiple track lines to cross and connect. Railway turnouts can transfer trains from the current track to another track according to operational needs. Any abnormality or malfunction in the RTS can severely disrupt train operations and potentially lead to major safety incidents, resulting in significant risks to personal safety and property [1]. Therefore, enhancing the reliability and safety of the RTS is crucial for ensuring operational safety, improving transportation efficiency, and reducing maintenance costs.

RTS has the characteristics of large quantity and complex construction and is one of the weak links in the railway transportation system. The railway switch machine is the pivotal execution component of the RTS that enables line switching. Some scholars and engineers have conducted much research on intelligent monitoring, reliability, and safety improvement of RTS or switch machines during operation. These studies can be roughly divided into two principal categories: data-driven and model-based methods [2], [3].

The data-driven approach does not require system mechanism modeling. It relies primarily on historical monitoring data from RTS, such as current, voltage, power, and force [4], [5]. By using machine learning or deep learning methods, a multilayer model is trained by extracting relevant features to construct a mapping relationship between the turnout system's operating and safety status [6]. Liu et al. [7] proposed an acoustic signal processing method based on multitimescale variational modal decomposition for trackside switch machine sound signals susceptible to noise by randomly simulating the real trackside noise and then training a support vector machine as a diagnostic model. Ji et al. [5] proposed a method based

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on segmenting critical points in the current profile for fault detection during switching, which reduces misclassification by suitable threshold selection. The effectiveness of the above machine learning methods depends on the quality of the data and the selection of features. With the continuous development of deep learning, a series of approaches based on deep self-encoders and deep neural networks has also been proposed [8]. Chen et al. [9] proposed a double-scale convolution neural network with wide first-layer kernels based on vibration signals with a low sampling rate for rutting machine fault diagnosis. However, deep learning models usually require a large amount of labeled data for training and do not reflect the operating mechanisms of actual physical systems. It prevents researchers and engineers from clearly understanding the model decision-making process and is not conducive to analyzing the impact of changes in operational status on the research object.

Model-based approaches typically use the system's input, output, and state variables, combined with physical prior knowledge, to construct a mathematical reference model capable of accurately describing the system's dynamic behavior. Based on this model, the accurate estimation of the state can provide a reliable basis for system monitoring and safety decisions. Márquez et al. [10] proposed an unobserved component model (UCM) established in a state-space framework for assessing wear in RTS, using a Kalman filter (KF) with a fixed-interval smoothing algorithm for state estimation. Wang et al. [11] proposed a weather-related fault prediction model based on Bayesian networks for fault diagnosis and prevention of railroad turnout systems. Still, the method relies on a large amount of historical data, especially weather data, to obtain accurate model parameters. In the RTS modeling process mentioned above, only the input and output signals of the system were utilized without incorporating any physical knowledge of the system. Consequently, the constructed models do not directly describe the real RTS. Hamadache et al. [12] established a simplified model using the relationship between the rotational displacement of the ballscrew and the stator current of the motor in the switch machine. They proposed a cumulative residual fault diagnosis method based on the cumulative residual fault diagnosis method, which realizes the fault diagnosis of the excessive friction or resistance in RTS [12]. Similarly, Dutta et al. [13] used the same approach to the RTS mathematical model. These methods overlook the impact of numerous nonlinear factors during the modeling process, resulting in a model that significantly deviates from the actual turnout system. This discrepancy limits the practical application of research aimed at enhancing RTS safety based on these modeling methods.

With the rapid development of high-speed railroads, the safety requirements for railway systems have become increasingly stringent. Precise state estimation of the RTS becomes more critical, as any deviation can lead to severe consequences. Accurate state estimation relies on highly accurate system models, making precise modeling of RTS a crucial issue that needs to be addressed in current model-based approaches. The RTS is typically a complex assembly of multiple components and subsystems, such as drivers, controllers, sensors, and actuators [14]. There is some nonlinear coupling of these components. Furthermore, RTS operates in outdoor environments and is subject to various disturbances, including vibrations from passing trains, geological changes, and weather fluctuations. These disturbances can cause mechanical damage and contact anomalies, leading to nonlinear effects such as changes in friction conditions and gear backlash. In the modeling process, it is essential to represent these nonlinear characteristics adequately. Additionally, depending on the type of switch machine used, the RTS may involve different engineering disciplines, including electrical, mechanical, and hydraulic engineering. Therefore, appropriate methods should be selected to address these specific characteristics in the modeling work.

The bond graph (BG) is a graphical representation method used to describe the structure and behavior of complex systems. It provides a unified and systematic approach for modeling and analyzing energy flows within complex systems and has been widely applied in engineering fields such as mechanics, electronics, and thermodynamics [15], [16]. Kumar et al. [17] proposed a functional analysis method for redundant mechatronic systems based on BG modeling theory for robot fault diagnosis. Amirdehi et al. [18] employed the BG method to model a train traction chain containing nonlinear transformers, accurately simulating the motion characteristics of the traction chain. Ladj et al. [19] applied the BG method to model RTS and compared the diagnostic effectiveness of data-driven methods versus model-based approaches. However, their models did not account for the impact of nonlinear factors within the RTS, resulting in a lack of accuracy. Therefore, this study undertakes nonlinear system modeling based on BG theory to achieve a more precise model of the RTS.

Accurate models are instrumental in conducting high-precision estimations of the operating status of RTS. KF combines system models and measurement data to design an optimal filter based on the criterion of minimizing mean squared error (mse) [20]. However, it is limited to linear Gaussian models [21]. For nonlinear systems, suboptimal estimation methods are required for approximation. Techniques such as the extended KF (EKF) [22], which uses analytical approximation, and the unscented KF (UKF) [23], which employs sampling approximation, can be applied to state estimation with moderate nonlinearity. The EKF uses the linear term of the Taylor series expansion, ignoring second-order and higher order terms. The UKF's parameter selection can lead to a nonpositive definite covariance matrix, making the algorithm challenging to execute [24]. Building on these methods, the adaptive EKF (AEKF) and adaptive UKF (AUKF) have been proposed. Their basic idea is to iteratively update the noise matrix adaptively to improve the estimation accuracy for nonlinear systems [25]. Most subsequent improvements to filtering methods extend these approaches or combine them with the characteristics of the application object. Although these improved methods can achieve better estimation accuracy, they still do not fully utilize the higher order terms of nonlinear systems, leaving room for further enhancement in filtering performance.

TABLE I
GENERALIZED VARIABLES IN DIFFERENT ENERGY DOMAINS

Domain	Effort e	Flow f	Generalized momentum p	Generalized displacement q
Electrical	Voltage (V)	Current (A)	Magnetic flux ($\mathrm{V \cdot s}$)	Charge (C)
Mechanical rotation	Torque ($\mathrm{N \cdot m}$)	Angular velocity ($\mathrm{rad/s}$)	Angular momentum ($\mathrm{N \cdot m \cdot s}$)	Angular displacement (rad)
Mechanical translation	Force (N)	Velocity ($\mathrm{m/s}$)	Momentum ($\mathrm{N \cdot s}$)	Displacement (m)
Hydraulics	Pressure (Pa)	Volume flow rate ($\mathrm{m^3/s}$)	Pressure momentum ($\mathrm{Pa \cdot s}$)	Volume ($\mathrm{m^3}$)

High-precision state estimation of nonlinear RTS models enables the early identification of potential faults and optimization of maintenance strategies and enhances the system's safety and reliability. It ensures the efficient and stable operation of turnout systems in complex environments.

The main work of this article is to develop a nonlinear RTS model and study its high-precision state estimation methods. The main contributions are summarized as follows.

1) Considering the influence of nonlinear friction and gear backlash on the system's dynamic characteristics, a precise nonlinear model of the RTS is established using BG unified modeling theory.
2) An improved EKF method, which incorporates second- and third-order statistical information from the Taylor series expansion, is proposed and applied to the high-precision state estimation of the RTS.
3) The impact of nonlinear factors on model accuracy was analyzed through experiments, and the superiority of the proposed filtering method was validated.

The remainder of this article is organized as follows. In Section II, the nonlinear modeling process of RTS is presented. Section III gives the design process for the improved filter. Section IV provides validation experiments to reflect the excellent performance of the proposed approach, and conclusions are described in Section V.

II. RTS MODELING

A. BG Modeling Theory

BG is a graphical dynamic modeling technique based on energy conservation, grouping system dynamics into four generalized variables: effort e, flow f, generalized momentum p, and generalized displacement q. Among them, the product of e and f represents power P, termed power variables, while p and q, as the integrals of e and f, describe energy, termed energy variables. Table I summarizes these variables across mechanical, electrical, and hydraulic domains.

The primary elements of BG modeling include the effort source Se, flow source Sf, dissipative element R, inertial element I, capacitive element C, transformer TF, gyrator GY, and junctions 0 and 1. The elements Se and Sf represent energy sources, such as voltage or current sources, with their modulated counterparts (MSe or MSf) being signal-controlled. The passive elements R, I, and C are single-port components: R models energy dissipation phenomena (e.g., resistance and friction), I represents the kinetic energy storage (e.g., inductance and inertia), and C stores power in the form of potential energy (e.g., capacitors and springs).

The two-port elements TF (transformer) and GY (gyrator) are used for energy conversion. TF models proportional conversions within the same domain (e.g., gears and levers), while GY describes cross-domain conversions (e.g., electric motors and hydraulic pumps). Junctions 0 and 1 facilitate power distribution in the system: in junction 0, flow variables are equal, and effort variables sum to zero, whereas in junction 1, effort variables are equal, and flow variables sum to zero.

The components are connected by connecting wires called bonds. Half arrows are used on the bond to indicate the direction of power flow, and short lines perpendicular to the bond are used to describe causality. In addition, information bonds can be used to indicate the transmission of signals from sensors, controllers, etc., which are connecting lines with complete arrows.

In the BG, junction 0, junction 1, and causality all provide constraint relationships for the modeling of the system. Based on the above characteristics, the BG modeling method has been widely used in the modeling and simulating of complex systems in multidisciplinary fields such as mechanics, electricity, fluid, thermodynamics, and control.

The BG method offers unique advantages in modeling RTS. It facilitates the simultaneous representation of coupling relationships across multiple energy domains, including mechanical, electrical, and sensor components, thereby enhancing modeling efficiency and providing a more intuitive modeling process. Additionally, the BG approach is highly extensible, allowing for the flexible addition of elements such as impedance components, energy storage devices, and sensors. It also supports the replacement of constitutive equations to transition from linear to nonlinear models quickly. Furthermore, its modular nature enables seamless adaptation to different types of switch machines. These features make the BG method an efficient and accurate tool for developing RTS models based on physical mechanisms.

B. RTS Modeling Process

This study modeled a single-machine traction RTS driven by a ZD-A-type dc electric switch machine, which can be extendable to other types of dc electric switch machines. This section gradually establishes the BG model by analyzing the main modules of this RTS, considering nonlinear factors, and deriving the state-space model. The switch machine, as the core actuator of the turnout system, not only directly determines its operational state but also exhibits complex nonlinear characteristics, with its dynamic behavior critically influencing the system's overall performance. The schematic of the ZD6-A switch machine and its main modules are shown in Fig. 1. On the right side of the figure is a schematic of the equivalent model connections of the critical components.

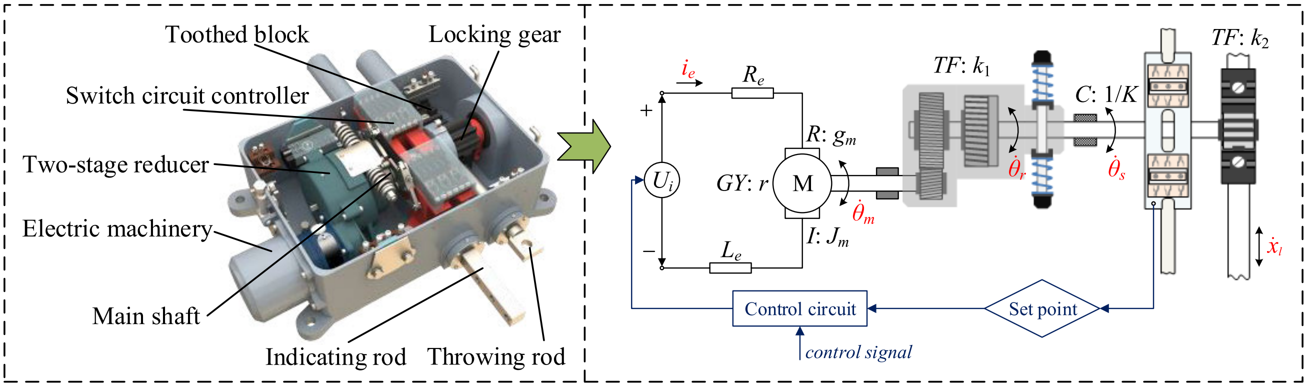


Fig. 1. Schematic of ZD6-A-type switch machine.

There are five variables marked in red, i_e, $\dot{\theta}_m$, $\dot{\theta}_r$, $\dot{\theta}_s$, and $\dot{x}_l$, which represent the flow variables at different parts: motor current, motor angular velocity, two-stage reducer output angular velocity, spindle angular velocity, and throwing rod velocity, respectively.

The critical operating components of this switch machine can be summarized as the motor electrical part, the motor mechanical part, the two-stage reducer, the main shaft, the locking gears, and the toothed block. The switch rail is connected to the toothed block in the turnout as a load through the throwing rod. Meanwhile, the indicating rod moves with the switch rail to the specified position. Then, it pushes the switch circuit controller to realize the motor power supply's interruption to complete the turnout's switching action. The process can be described by constructing a word BG, i.e., using words to represent subsystems or components connected by bonds, as shown in Fig. 2(a).

Real RTS systems exhibit various nonlinear factors, with this study focusing on two typical nonlinearities: gear backlash and nonlinear friction. The gear backlash model is developed based on the effects of clearance on torque transmission and has undergone rigorous stability validation [26]. Nonlinear friction is described using the Stribeck friction model, which captures variations in friction coefficients at different sliding speeds while highlighting the influence of lubrication. Proper lubrication of moving components, such as those in the switch machine and between the switch rail and base plate, is crucial in turnout system maintenance. An RTS BG model can be constructed based on the word BG using the following procedure.

1) Electric Machinery Drive Part: The control signal and switch circuit controller activate and deactivate the supply voltage. The operation of the switch circuit controller is realized by moving the indicating rod to the set point position.

2) Electrical Part of the Motor: The equivalent circuit model of a dc motor is given in Fig. 1, which contains the equivalent internal resistance R_e and inductance L_e. The flow sensor $\mathrm{Df}: i_e$ measures the current of the electric machinery. The bonding diagram element $\mathrm{GY}: r$ represents the power conversion from the electrical part to the mechanical part of the motor, where r is the motor torque constant.

3) Mechanical Part of the Motor: The element $I{:}J_m$ represents the rotational moment of inertia of the motor and $\mathrm{Df}: \dot{\theta}_m$ describes the measurement of the angular velocity of the motor. The element $R : R_m$ of the mechanical part is constructed using the Stribeck nonlinear friction model as

$$\mathcal{F}_1(\dot{\theta}_m, T_m) = \begin{cases} T_m, & \dot{\theta}_m = 0 \,\&\, |T_m| < f_{ms} \\ f_{ms} \cdot \mathrm{sgn}(T_m), & \dot{\theta}_m = 0 \,\&\, |T_m| > f_{ms} \\ b_m(\dot{\theta}_m) & \text{otherwise} \end{cases} \tag{1}$$

where T_m is the electromagnetic torque of the motor, f_{ms} is the moment of static friction, and $b_m(\dot{\theta}_m)$ denotes the Stribeck friction curve, which can usually be expressed as

$$\begin{aligned} b_m(\dot{\theta}_m) =& \lambda_m \cdot \dot{\theta}_m + \mathrm{sgn}(\dot{\theta}_m) \\ & \cdot \left[f_{mc} + (f_{ms} - f_{mc}) \cdot e^{-\frac{|\dot{\theta}_m|}{\sigma_1}} \right] \end{aligned} \tag{2}$$

where f_{mc} is the Coulomb friction moment. λ_m is the coefficient of viscous friction, representing the relationship between the magnitude of frictional force and sliding velocity under lubricated conditions. The σ_1 denotes the Stribeck friction velocity threshold, which is the critical velocity at which the friction force transitions from static friction to dynamic friction.

In addition, this section should consider the interference of the gear backlash phenomenon on the torque transmission, and the propagation of the motion generates a corresponding hysteresis according to the backlash size. Based on the dead zone generated by the backlash and the rebound mechanism generated by the gear collision, a continuously conductible nonlinear model of the gear backlash is established as follows:

$$\begin{cases} T_r = \dfrac{T_m}{k_1} + d_m \\ d_m = -4K\varphi \dfrac{1 - e^{-\frac{1}{2\varphi}\Delta\theta}}{1 + e^{-\frac{1}{2\varphi}\Delta\theta}} \end{cases} \tag{3}$$

where T_r is the output torque of the two-stage gearbox, K is the stiffness, and φ is the backlash amplitude (in rad). Define $\Delta\theta = \theta_m - \theta_r/k_1$ as the angular displacement difference. Here, θ_m and θ_r represent the angular displacements of the input and output parts of the reducer, respectively. The d_m is the disturbance moment generated by the backlash, which can be expressed in terms of modulated sources $\mathrm{MSe}: d_m$.

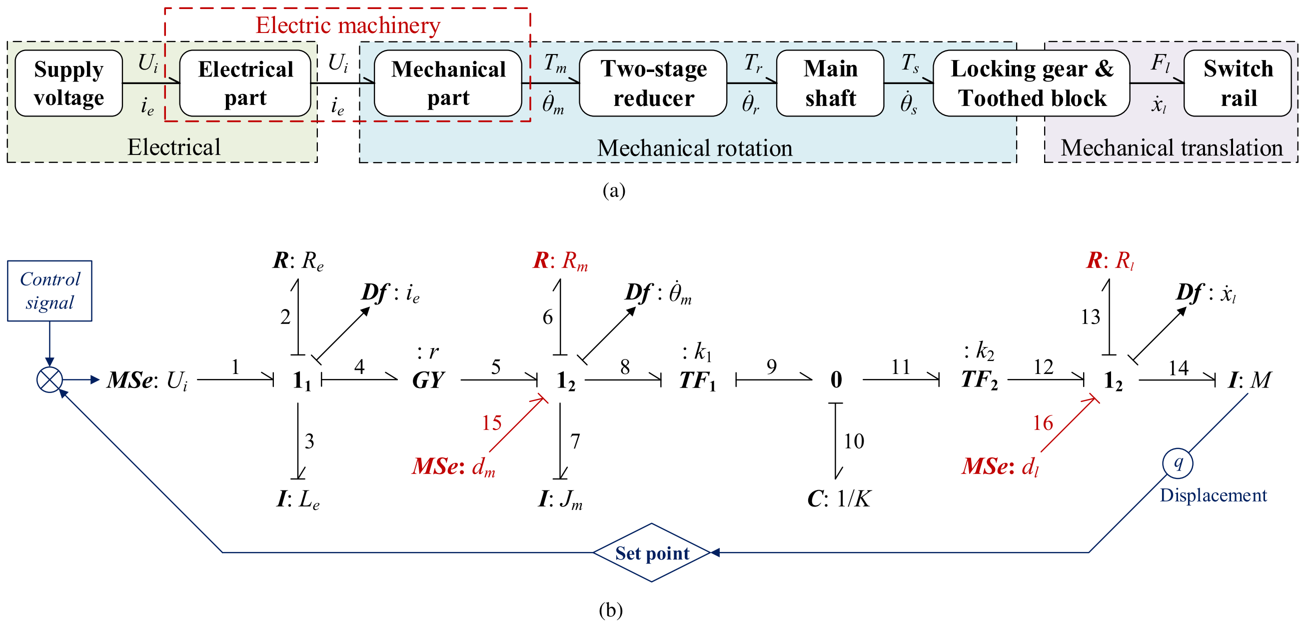


Fig. 2. Model construction of RTS. (a) Word BG of RTS. (b) BG model of RTS.

4) Two-Stage Reducer Part: In a ZD6-A-type switch machine, the first stage reducer is an external gear reducer, the second stage reducer is an internal gear planetary reducer, and the total reduction ratio of the two stages is k_1, which can be expressed by the bonding diagram element TF1 : k_1.

5) Main Shaft Part: The main shaft, which connects the reducer's output to the locking gear, can be represented by the capacitive element $C : 1/K$, where K is the stiffness of the main shaft.

6) Locking Gear, Toothed Block, and Switch Rail: The main shaft drives the locking gears and converts mechanical rotary motion to mechanical translational motion via the toothed block. This conversion ratio is k_2 and can be represented by the bonding diagram element TF1 : k_2. Similarly, concerning the mechanical portion of the motor, the element $R : R_l$ is constructed utilizing the Stribeck nonlinear friction model

$$\mathcal{F}_2(\dot{x}_l, T_s) = \begin{cases} T_s, & \dot{x}_l = 0 \& |T_s| < f_{ls} \\ f_{ls} \cdot \operatorname{sgn}(T_s), & \dot{x}_l = 0 \& |T_s| > f_{ls} \\ b_l(\dot{x}_l), & \text{otherwise} \end{cases} \quad (4)$$

with

$$b_l(\dot{x}_l) = \lambda_l \cdot \dot{x}_l + \operatorname{sgn}(\dot{x}_l) \cdot \left[f_{lc} + (f_{ls} - f_{lc}) \cdot e^{-\frac{|\dot{x}_l|}{\sigma_2}} \right] \quad (5)$$

where T_s is the torque provided by the main shaft, and $b_l(\dot{x}_l)$ denotes the Stribeck friction curve. As well, f_{ls}, λ_l, f_{lc}, and σ_2 are, respectively, the static friction moment, the coefficient of viscous friction, the Coulomb friction moment, and the velocity threshold for this section. The backlash disturbance torque in this section is given by the element MSe : d_l, and according to the torque transfer process, it can be shown as

$$d_l = k_1 k_2 d_m = -4 k_1 k_2 K \varphi \frac{1 - e^{-\frac{1}{2\varphi}\Delta\theta}}{1 + e^{-\frac{1}{2\varphi}\Delta\theta}}. \quad (6)$$

Considering the switch rail as a loaded part, it can be represented by the inertial element $I : M$, where M is the load mass.

The nonlinear BG model of the ZD6-A single traction RTS can be derived from the aforementioned modeling process, as shown in Fig. 2(b).

In order to verify the accuracy of the model, the BG model is constructed using the simulation software "20-SIM" and combined with the real collected current data of a ZD6 switch machine as a comparison, as shown in Fig. 3. Table II gives the settings of the main parameters in the BG model. The parameter values were primarily determined based on product manuals, relevant literature, and simulation experiments, reflecting the operating conditions of the equipment under normal working scenarios.

The BG model current shown in Fig. 3 closely aligns with the actual current, effectively capturing the core dynamics of turnout conversion. The displacement curve at the load (the connection point between the switch rail and the throwing rod) derived from the model indicates a switch machine travel output of 163.44 mm. This result falls within the calibrated range of the ZD6-A switch machine travel (165 $\pm$ 2 mm), demonstrating the model's precision. Furthermore, the simulated switch machine conversion time is 3.76 s, consistent with its nominal value of $\leq$3.8 s. These findings confirm the high accuracy of the nonlinear RTS model developed in this study.

C. State-Space Model Derivation

The state-space model can describe the system's dynamic behavior in mathematical equations between inputs, outputs, and state variables, making it easy to simulate, analyze, and estimate the state. The state-space equations describing the RTS can be easily derived from the BG model.

TABLE II
PARAMETERS IN THE RTS-BG MODEL

Parameter	Value	Parameter	Value
U_i	160 V	R_e	10 Ω
L_e	0.0011 (H)	r	0.875
J_m	0.0025 ($\mathrm{Kg}\cdot\mathrm{m}^2$)	K	10^{-2} $\mathrm{N}\cdot\mathrm{m/rad}$
k_1	0.0064	k_2	0.05
λ_m	0.03	f_{ms}	0.025 $\mathrm{N}\cdot\mathrm{m}$
f_{mc}	0.02 $\mathrm{N}\cdot\mathrm{m}$	σ_1	0.02
λ_l	0.06	f_{ls}	0.4 $\mathrm{N}\cdot\mathrm{m}$
f_{lc}	0.35 $\mathrm{N}\cdot\mathrm{m}$	σ_2	0.02
φ	0.025 rad	M	380 Kg

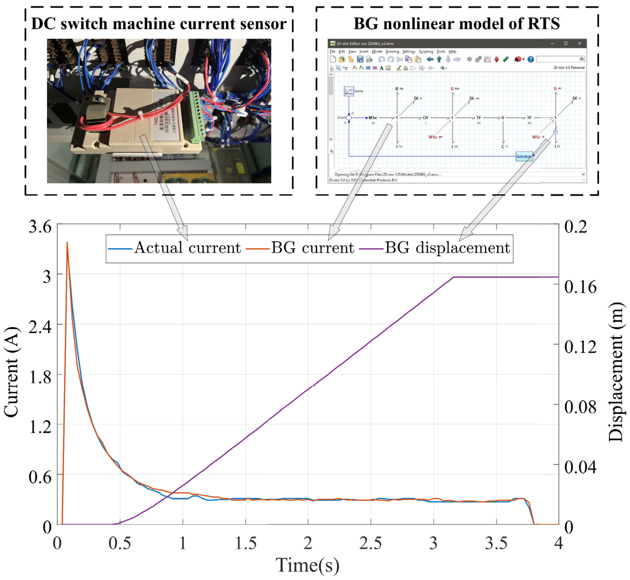


Fig. 3. BG simulation and model validation.

For junction 1_1, the sum of effort variables is zero, as

$$e_1 - e_2 - e_3 - e_4 = 0 \tag{7}$$

and

$$\begin{cases} e_1 = U_i \\ e_2 = i_e R_e \\ e_3 = L_e \dfrac{di_e}{dt} \\ e_4 = r f_5 = r\dot{\theta}_m. \end{cases} \tag{8}$$

Substituting (8) into (7) yields the state equation for the electrical part of the dc motor in the switch machine as follows:

$$\frac{di_e}{dt} = -\frac{R_e}{L_e} i_e - \frac{r}{L_e}\dot{\theta}_m + \frac{1}{L_e} U_i. \tag{9}$$

For junction 1_2, there is likewise

$$e_5 + d_m - e_6 - e_7 - e_8 = 0 \tag{10}$$

and

$$\begin{cases} e_5 = r i_e \\ e_6 = b_m(\dot{\theta}_m) \\ e_7 = J_m \dfrac{d\dot{\theta}_m}{dt} \\ e_8 = k_1 e_9 = k_1 e_{10} = k_1 K \displaystyle\int f_{10} dt. \end{cases} \tag{11}$$

On junction 0, the sum of all flow variables is zero, i.e., $f_9 - f_{10} - f_{11} = 0$, so the above equation has

$$\begin{aligned} e_8 &= k_1 K \int (f_9 - f_{11}) dt = k_1 K \int \left(k_1 f_8 - \frac{1}{k_2} f_{12}\right) dt \\ &= k_1 K \left(k_1 \theta_m - \frac{1}{k_2} x_l\right). \end{aligned} \tag{12}$$

Substituting (11) and (12) into (10) yields the state equation for the mechanical part of the switch machine motor and reducer as follows:

$$\begin{aligned} \frac{d\dot{\theta}_m}{dt} = &\frac{r}{J_m} i_e + \frac{1}{J_m} d_m - \frac{1}{J_m} b_m(\dot{\theta}_m) \\ &- \frac{k_1 K}{J_m}\left(k_1 \theta_m - \frac{1}{k_2} x_l\right). \end{aligned} \tag{13}$$

For junction 1_3, similarly, there is

$$e_{12} + d_l - e_{13} - e_{14} = 0 \tag{14}$$

and

$$\begin{cases} e_{12} = \dfrac{1}{k_2} e_{11} = \dfrac{1}{k_2} e_{10} = \dfrac{K}{k_2}\left(k_1\theta_m - \dfrac{1}{k_2} x_l\right) \\ e_{13} = b_l(\dot{x}_l) \\ e_{14} = M \dfrac{d\dot{x}_l}{dt}. \end{cases} \tag{15}$$

Substituting (14) and (15) into (13) yields the state-space equations for the locking gear, toothed block, and turnout load as follows:

$$\frac{d\dot{x}_l}{dt} = \frac{K}{M k_2}\left(k_1\theta_m - \frac{1}{k_2} x_l\right) + \frac{1}{M} d_l - \frac{1}{M} b_l(\dot{x}_l). \tag{16}$$

Combining (9), (13), and (16), the state-space nonlinear model of the RTS can be obtained as

$$\begin{cases} \dfrac{di_e}{dt} = -\dfrac{R_e}{L_e} i_e - \dfrac{r}{L_e}\dot{\theta}_m + \dfrac{1}{L_e} U_i \\ \dfrac{d\dot{\theta}_m}{dt} = \dfrac{r}{J_m} i_e + \dfrac{1}{J_m} d_m - \dfrac{1}{J_m} b_m(\dot{\theta}_m) \\ \qquad\quad - \dfrac{k_1 K}{J_m}\left(k_1\theta_m - \dfrac{1}{k_2} x_l\right) \\ \dfrac{d\dot{x}_l}{dt} = \dfrac{K}{M k_2}\left(k_1\theta_m - \dfrac{1}{k_2} x_l\right) + \dfrac{1}{M} d_l - \dfrac{1}{M} b_l(\dot{x}_l). \end{cases} \tag{17}$$

III. HIGH-PRECISION FILTER DESIGN

KF is a method that achieves recursive optimal estimation in Gaussian noisy linear systems. For certain nonlinear systems, the EKF approximates the nonlinearized model by performing a Taylor series expansion of the nonlinear function and then uses the KF for estimation. However, the EKF linearization process introduces a large estimation error in systems with a high degree of nonlinearity since only the primary term is considered, leading to filter performance degradation. This section proposes an improved EKF method that considers the higher order terms of the nonlinear function expansion for high-precision estimation of the RTS operating state.

Discretizing (17) and assuming that it conforms to a Gaussian distribution, the general form of the nonlinear state model of the RTS can be notated as

$$\boldsymbol{x}(k+1)=\boldsymbol{\phi}(\boldsymbol{x}(k))+\boldsymbol{B}(k)\boldsymbol{u}(k)+\boldsymbol{w}(k) \tag{18}$$

where $\phi(\cdot)$ is the nonlinear state transfer function, $\boldsymbol{x}(k)\in\mathbb{R}^n$ is the system state vector, $\boldsymbol{B}(k)$ is the system input matrix, $\boldsymbol{u}(k)\in\mathbb{R}^r$ is the system input vector, $\boldsymbol{y}(k)\in\mathbb{R}^m$ is a vector of measured values, and $\boldsymbol{w}(k)\sim N(0,\boldsymbol{Q}(k))$ is the modeling error.

Based on the setup of the flow variable sensor in the BG model of the RTS, the linear measurement model can be obtained as

$$\boldsymbol{y}(k+1)=\boldsymbol{H}(k+1)\boldsymbol{x}(k+1)+\boldsymbol{v}(k+1) \tag{19}$$

where $\boldsymbol{H}(k+1)$ is the measurement matrix, and $\boldsymbol{v}(k+1)\sim N(0,\boldsymbol{R}(k+1))$ is the measurement noise.

Similar equations in the form of (18) and (19) constitute a class of discrete-time nonlinear state-space models consisting of nonlinear state equations and linear measurement equations

$$\begin{cases}\boldsymbol{x}(k+1)=\boldsymbol{\phi}(x(k))+\boldsymbol{B}(k)\boldsymbol{u}(k)+\boldsymbol{w}(k)\\ \boldsymbol{y}(k+1)=\boldsymbol{H}(k+1)\boldsymbol{x}(k+1)+\boldsymbol{v}(k+1)\end{cases} \tag{20}$$

where $\phi(\cdot)$ can be considered a state transfer function with continuously differentiable properties of arbitrary order. The initial state $\boldsymbol{x}(0)\sim N(\hat{\boldsymbol{x}}_0,\boldsymbol{P}_0)$, system noise $\boldsymbol{w}(k)$, and measurement noise $\boldsymbol{v}(k+1)$ are statistically independent of each other. Denote $\hat{\boldsymbol{x}}(k|k)$ and $\boldsymbol{P}(k|k)$ as the estimated value of the state and the estimation error covariance matrix at the moment k, respectively. In order to realize high-precision state estimation for such systems, this article proposes an EKF design method based on the statistical features of the third-order Taylor expansion.

A. Acquisition of Second- and Third-Order Statistical Information

In the third-order Taylor expansion, both third-order statistical information and second-order statistical information are included. The third-order Taylor expansion $\boldsymbol{x}_{\{3\}}(k+1)$ of the nonlinear model (18) at $\hat{\boldsymbol{x}}(k|k)$ can be obtained as follows:

$$\begin{aligned}\boldsymbol{x}_{\{3\}}(k+1)=\boldsymbol{\phi}(\hat{\boldsymbol{x}}^{(1)}(k|k))+\sum_{i=1}^{3}\boldsymbol{A}^{(i)}(k)\tilde{\boldsymbol{x}}^{(i)}(k|k)\\ +\boldsymbol{B}(k)\boldsymbol{u}(k)+\boldsymbol{w}(k)\end{aligned} \tag{21}$$

with

$$\boldsymbol{A}^{(i)}(k)=\frac{1}{i!}\frac{\partial^i\boldsymbol{\phi}(\boldsymbol{x}(k))}{\partial\boldsymbol{x}(k)^i}\bigg|_{\boldsymbol{x}(k)=\hat{\boldsymbol{x}}(k|k)} \tag{22}$$

where $\tilde{\boldsymbol{x}}^{(i)}(k|k)$ represents the ith-order Kronecker product of the original estimation error $\tilde{\boldsymbol{x}}(k|k)$, whose estimate and estimation error covariance matrices are, respectively, denoted as

$$\begin{aligned}\hat{\tilde{\boldsymbol{x}}}^{(i)}(k|k)&=E\left\{\tilde{\boldsymbol{x}}^{(i)}(k|k)\right\}\\ P_{\tilde{x}\tilde{x}}{}^{(i,i)}(k|k)&=E\left\{[\tilde{x}^{(i)}(k|k)-\hat{\tilde{x}}^{(i)}(k|k)]\right.\\ &\qquad\left.\cdot[\tilde{x}^{(i)}(k|k)-\hat{\tilde{x}}^{(i)}(k|k)]^{\mathrm{T}}\right\}.\end{aligned}$$

Bringing (21) into the measurement equation of (20), then

$$\begin{aligned}\boldsymbol{y}_{\{3\}}(k+1)&=\boldsymbol{H}(k+1)\boldsymbol{x}_{\{3\}}(k+1)+\boldsymbol{v}(k+1)\\ &=\boldsymbol{H}(k+1)\boldsymbol{\phi}(\hat{\boldsymbol{x}}^{(1)}(k|k))\\ &\quad+\boldsymbol{H}(k+1)\boldsymbol{A}^{(1)}(k)\tilde{\boldsymbol{x}}^{(1)}(k|k)\\ &\quad+\boldsymbol{H}(k+1)\boldsymbol{A}^{(2)}(k)\tilde{\boldsymbol{x}}^{(2)}(k|k)\\ &\quad+\boldsymbol{H}(k+1)\boldsymbol{A}^{(3)}(k)\tilde{\boldsymbol{x}}^{(3)}(k|k)\\ &\quad+\boldsymbol{H}(k+1)\boldsymbol{B}(k)\boldsymbol{u}(k)\\ &\quad+\boldsymbol{H}(k+1)\boldsymbol{w}(k)+\boldsymbol{v}(k+1.\end{aligned} \tag{23}$$

By organizing this equation and making $\tilde{\boldsymbol{X}}_{\{3\}}(k|k)=[\tilde{\boldsymbol{x}}^{(2)}(k|k),\tilde{\boldsymbol{x}}^{(3)}(k|k)]^{\mathrm{T}}$, the new measurement equation for $\tilde{\boldsymbol{x}}^{(2)}(k|k)$ and $\tilde{\boldsymbol{x}}^{(3)}(k|k)$ is obtained as

$$\boldsymbol{Y}_{\{3\}}(k+1)=\boldsymbol{\Gamma}_{\{3\}}(k+1)\tilde{\boldsymbol{X}}_{\{3\}}(k|k)+\boldsymbol{V}_{\{3\}}(k+1) \tag{24}$$

which

$$\begin{cases}\boldsymbol{Y}_{\{3\}}(k+1)=\boldsymbol{y}_{\{3\}}(k+1)-\boldsymbol{H}(k+1)\boldsymbol{\phi}(\hat{\boldsymbol{x}}^{(1)}(k|k))\\ \qquad\qquad-\boldsymbol{H}(k+1)\boldsymbol{B}(k)\boldsymbol{u}(k)\\ \boldsymbol{\Gamma}_{\{3\}}(k+1)=\left[\boldsymbol{H}(k+1)\boldsymbol{A}^{(2)}(k),\boldsymbol{H}(k+1)\boldsymbol{A}^{(3)}(k)\right]\\ \boldsymbol{V}_{\{3\}}(k+1)=\boldsymbol{H}(k+1)\boldsymbol{A}^{(1)}(k)\tilde{\boldsymbol{x}}^{(1)}(k|k)\\ \qquad\qquad+\boldsymbol{H}(k+1)\boldsymbol{w}(k)+\boldsymbol{v}(k+1).\end{cases} \tag{25}$$

In the new measurement equation (24), the statistical properties of $\boldsymbol{V}_{\{3\}}(k+1)$ satisfy $\boldsymbol{V}_{\{3\}}(k+1)\sim N(0,\boldsymbol{R}_{\{3\}}(k+1))$ and

$$\begin{aligned}E\{\boldsymbol{V}_{\{3\}}(k+1)\}&=\boldsymbol{H}(k+1)\boldsymbol{A}^{(1)}E\{\tilde{\boldsymbol{x}}(k|k)\}\\ &\quad+\boldsymbol{H}(k+1)E\{\boldsymbol{w}(k)\}+E\{\boldsymbol{v}(k+1)\}\\ &=0\end{aligned} \tag{26}$$

$$\begin{aligned}\boldsymbol{R}_{\{3\}}(k+1)&=E\{\boldsymbol{V}_{\{3\}}(k+1)\boldsymbol{V}_{\{3\}}(k+1)^{\mathrm{T}}\}\\ &=\boldsymbol{H}(k+1)\boldsymbol{A}^{(1)}(k)\boldsymbol{P}(k|k)\boldsymbol{A}^{(1)}(k)^{\mathrm{T}}\boldsymbol{H}(k+1)^{\mathrm{T}}\\ &\quad+\boldsymbol{H}(k+1)\boldsymbol{Q}(k)\boldsymbol{H}(k+1)^{\mathrm{T}}+\boldsymbol{R}(k+1).\end{aligned} \tag{27}$$

The new linear measurement equation represented by using (24), where the number of unknown variables exceeds the available equations, can be solved for $\tilde{\boldsymbol{X}}_{\{3\}}(k|k)$ by using the underdetermined least-squares (ULS) method as [27]

$$\begin{aligned}\hat{\tilde{\boldsymbol{X}}}_{\{3\}}(k|k)&=\boldsymbol{\Gamma}_{\{3\}}(k+1)^{\mathrm{T}}\left[\boldsymbol{\Gamma}_{\{3\}}(k+1)\cdot\boldsymbol{\Gamma}_{\{3\}}(k+1)^{\mathrm{T}}\right]^{-1}\\ &\quad\times\boldsymbol{Y}_{\{3\}}(k+1).\end{aligned} \tag{28}$$

The estimation error of $\tilde{\boldsymbol{X}}_{\{3\}}(k|k)$ is

$$\begin{aligned}\tilde{\tilde{\boldsymbol{X}}}_{\{3\}}(k|k)&=-\boldsymbol{\Gamma}_{\{3\}}(k+1)^{\mathrm{T}}\left[\boldsymbol{\Gamma}_{\{3\}}(k+1)\cdot\boldsymbol{\Gamma}_{\{3\}}(k+1)^{\mathrm{T}}\right]^{-1}\\ &\quad\times\boldsymbol{V}_{\{3\}}(k+1).\end{aligned} \tag{29}$$

Its estimation error covariance matrix is

$$\begin{aligned}\boldsymbol{P}_{\tilde{\tilde{X}}_{\{3\}}}(k|k)&=E\left\{\tilde{\tilde{\boldsymbol{X}}}_{\{3\}}(k|k)\tilde{\tilde{\boldsymbol{X}}}_{\{3\}}(k|k)^{\mathrm{T}}\right\}\\ &=\boldsymbol{\Gamma}_{\{3\}}(k+1)^{\mathrm{T}}\left[\boldsymbol{\Gamma}_{\{3\}}(k+1)\boldsymbol{\Gamma}_{\{3\}}(k+1)^{\mathrm{T}}\right]^{-1}\\ &\quad\cdot\boldsymbol{R}_{\{3\}}(k+1)\left[\boldsymbol{\Gamma}_{\{3\}}(k+1)\boldsymbol{\Gamma}_{\{3\}}(k+1)^{\mathrm{T}}\right]^{-1}\\ &\quad\cdot\boldsymbol{\Gamma}_{\{3\}}(k+1).\end{aligned} \tag{30}$$

By solving $\tilde{\boldsymbol{X}}_{\{3\}}(k|k)$, the statistical information, i.e., $\tilde{\boldsymbol{x}}^{(2)}(k|k)\sim N(\hat{\tilde{\boldsymbol{x}}}^{(2)}(k|k),\boldsymbol{P}_{\tilde{\boldsymbol{x}}\tilde{\boldsymbol{x}}}{}^{(2,2)}(k|k))$ and $\tilde{\boldsymbol{x}}^{(3)}(k|k)\sim$

$N(\hat{\tilde{x}}^{(3)}(k|k), P_{\tilde{x}\tilde{x}}{}^{(3,3)}(k|k))$, can be derived from the original nonlinear state equation obtained through the third-order Taylor expansion.

B. Filter Design

The design process of the improved EKF based on the third-order statistical properties is as follows.

Step 1: The one-step state prediction and prediction error are as follows:

$$\begin{aligned}\hat{\boldsymbol{x}}_{\{3\}}(k+1|k) &= \boldsymbol{\phi}(\hat{\boldsymbol{x}}(k|k)) + \sum_{i=1}^{3} \boldsymbol{A}^{(i)}(k)\hat{\tilde{\boldsymbol{x}}}^{(i)}(k|k)\\ &\quad + \boldsymbol{B}(k)\boldsymbol{u}(k)\end{aligned} \tag{31}$$

$$\begin{aligned}\tilde{\boldsymbol{x}}_{\{3\}}(k+1|k) &= \boldsymbol{x}_{\{3\}}(k+1) - \hat{\boldsymbol{x}}_{\{3\}}(k+1|k)\\ &= \sum_{i=1}^{3} \boldsymbol{A}^{(i)}(k)[\tilde{\boldsymbol{x}}^{(i)}(k|k) - \hat{\tilde{\boldsymbol{x}}}^{(i)}(k|k)] + \boldsymbol{w}(k)\\ &= \sum_{i=1}^{3} \boldsymbol{A}^{(i)}(k)\tilde{\tilde{\boldsymbol{x}}}^{(i)}(k|k) + \boldsymbol{w}(k).\end{aligned} \tag{32}$$

Step 2: The one-step measurement prediction and its error are

$$\hat{\boldsymbol{y}}_{\{3\}}(k+1|k) = \boldsymbol{H}(k+1)\hat{\boldsymbol{x}}_{\{3\}}(k+1|k) \tag{33}$$

and

$$\begin{aligned}\tilde{\boldsymbol{y}}_{\{3\}}(k+1|k) &= \boldsymbol{y}_{\{3\}}(k+1) - \hat{\boldsymbol{y}}_{\{3\}}(k+1|k)\\ &= \boldsymbol{H}(k+1)\boldsymbol{x}_{\{3\}}(k+1) + \boldsymbol{v}(k+1)\\ &\quad - \boldsymbol{H}(k+1)\hat{\boldsymbol{x}}_{\{3\}}(k+1|k)\\ &= \boldsymbol{H}(k+1)\tilde{\boldsymbol{x}}_{\{3\}}(k+1) + v(k+1).\end{aligned} \tag{34}$$

Step 3: The error covariance matrix of the one-step state prediction estimate is as follows:

$$\begin{aligned}\boldsymbol{P}_{\tilde{x}_{\{3\}}\tilde{x}_{\{3\}}}(k+1|k) &= E\left\{\tilde{\boldsymbol{x}}_{\{3\}}(k+1|k)\tilde{\boldsymbol{x}}_{\{3\}}(k+1|k)^{\mathrm{T}}\right\}\\ &= E\left\{\left[\sum_{i=1}^{3} \boldsymbol{A}^{(i)}(k)\tilde{\tilde{\boldsymbol{x}}}^{(i)}(k|k) + \boldsymbol{w}(k)\right]\right.\\ &\qquad \left.\cdot\left[\sum_{i=1}^{3} \boldsymbol{A}^{(i)}(k)\tilde{\tilde{\boldsymbol{x}}}^{(i)}(k|k) + \boldsymbol{w}(k)\right]^{\mathrm{T}}\right\}\\ &= \sum_{i=1}^{3} \boldsymbol{A}^{(i)}(k)\boldsymbol{P}_{\tilde{x}\tilde{x}}{}^{(i,i)}(k|k)\boldsymbol{A}^{(i)}(k)^{\mathrm{T}}\\ &\quad + \boldsymbol{Q}(k).\end{aligned} \tag{35}$$

Step 4: The filter gain is calculated as

$$K_{\{3\}}(k+1) = \frac{\boldsymbol{P}_{\tilde{x}_{\{3\}}\tilde{x}_{\{3\}}}(k+1|k)\boldsymbol{H}(k+1)^{\mathrm{T}}}{\boldsymbol{H}(k+1)\boldsymbol{P}_{\tilde{x}_{\{3\}}\tilde{x}_{\{3\}}}(k+1|k)\boldsymbol{H}(k+1)^{\mathrm{T}} + \boldsymbol{R}(k+1)}. \tag{36}$$

In practical computations, matrix inversion may become unstable under high-condition numbers, which can be addressed using regularization techniques.

Step 5: State estimate update

$$\begin{aligned}\hat{\boldsymbol{x}}_{\{3\}}(k+1|k+1) &= \hat{\boldsymbol{x}}_{\{3\}}(k+1|k)\\ &\quad + K_{\{3\}}(k+1)\tilde{\boldsymbol{y}}_{\{3\}}(k+1|k).\end{aligned} \tag{37}$$

Step 6: Estimation error and covariance matrix update

$$\begin{aligned}\tilde{\boldsymbol{x}}_{\{3\}}(k+1|k+1) &= \boldsymbol{x}_{\{3\}}(k+1) - \hat{\boldsymbol{x}}_{\{3\}}(k+1|k+1)\\ &= \boldsymbol{x}_{\{3\}}(k+1) - \hat{\boldsymbol{x}}_{\{3\}}(k+1|k)\\ &\quad - K_{\{3\}}(k+1)\tilde{\boldsymbol{y}}_{\{3\}}(k+1|k)\\ &= \tilde{\boldsymbol{x}}_{\{3\}}(k+1|k)\\ &\quad - K_{\{3\}}(k+1)\tilde{\boldsymbol{y}}_{\{3\}}(k+1|k)\end{aligned} \tag{38}$$

and

$$\begin{aligned}&\boldsymbol{P}_{\tilde{x}_{\{3\}}\tilde{x}_{\{3\}}}(k+1|k+1)\\ &= E\left\{\tilde{\boldsymbol{x}}_{\{3\}}(k+1|k+1)\tilde{\boldsymbol{x}}_{\{3\}}(k+1|k+1)^{\mathrm{T}}\right\}\\ &= E\left\{\left[\tilde{\boldsymbol{x}}_{\{3\}}(k+1|k) - K_{\{3\}}(k+1)\tilde{\boldsymbol{y}}_{\{3\}}(k+1|k)\right]\right.\\ &\qquad \left.\cdot\left[\tilde{\boldsymbol{x}}_{\{3\}}(k+1|k) - K_{\{3\}}(k+1)\tilde{\boldsymbol{y}}_{\{3\}}(k+1|k)\right]^{\mathrm{T}}\right\}\\ &= \boldsymbol{P}_{\tilde{x}_{\{3\}}\tilde{x}_{\{3\}}}(k+1|k)\\ &\quad - K_{\{3\}}(k+1)\boldsymbol{P}_{\tilde{x}_{\{3\}}\tilde{x}_{\{3\}}}(k+1|k)\boldsymbol{H}(k+1).\end{aligned} \tag{39}$$

The design of a high-precision improved EKF based on the third-order feature information has been completed now. For the analysis of how the filter enhances its performance by leveraging higher order term information, refer to Appendix I. This filter can be used to estimate the state variables in the RTS nonlinear model with high accuracy. The design process of this filter can be further iterated to obtain filters of higher orders. Furthermore, while the computational complexity of the 3oEKF is higher than that of the EKF, its code execution time remains comparable to that of the UKF. More importantly, the 3oEKF demonstrates a significant improvement in estimation accuracy compared to both the EKF and UKF. For a detailed analysis of algorithm complexity, please refer to Appendix II. However, in the Taylor expansion, the coefficients diminish at higher orders, so the higher order terms contain progressively less information. When the nonlinearity of the model is not high enough, the statistical information of higher orders is more limited to improve the filter's performance. Consequently, this article focuses on the third-order approximation of the nonlinearity.

IV. Experiment and Results Analysis

In this section, the accuracy of the established RTS nonlinear model and the performance of the proposed improved EKF are analyzed through ablation experiments and validation experiments.

A. Model Accuracy Ablation Experiment

To verify the accuracy of the RTS nonlinear model, this experiment explores the impact of nonlinear friction and gear backlash on the model's precision through ablation analysis. The model is divided into the following four scenarios for experimentation.

1) *Case 1:* Neither gear backlash nor nonlinear friction is included; this represents the most simplified linear model.

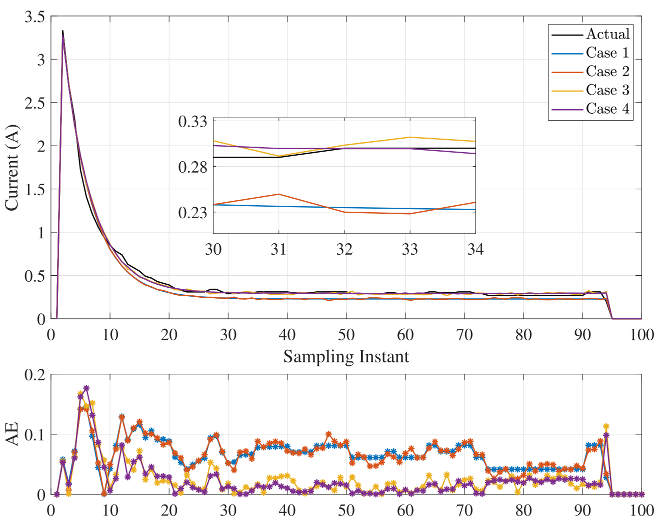


Fig. 4. Comparison between model output current and actual current in four cases.

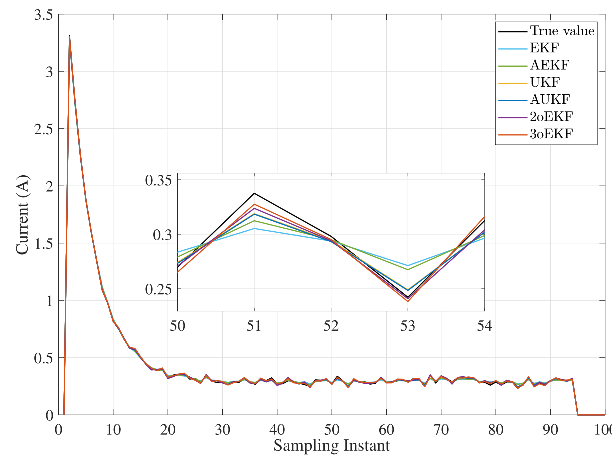


Fig. 5. Current state estimation under normal operating conditions.

TABLE III
COMPARISON OF MAE AND RMSE BETWEEN MODEL OUTPUT CURRENT AND ACTUAL CURRENT IN FOUR CASES

Cases	MAE	Improvement to Case 1	RMSE	Improvement to Case 1
Case 1	0.0620	/	0.0721	/
Case 2	0.0616	06.45%	0.0718	04.16%
Case 3	0.0314	49.35%	0.0456	36.75%
Case 4	0.0304	50.97%	0.0446	38.14%

2) *Case 2:* Only gear backlash is included, without considering the effect of nonlinear friction.
3) *Case 3:* Only nonlinear friction is included, without considering the effect of gear backlash.
4) *Case 4:* Both gear backlash and nonlinear friction are included.

The current curves generated by the model under the four cases, along with the actual collected current, are shown in Fig. 4. The figure also shows the absolute error (AE) between each point of the four current curves and the actual current. Comparative analysis reveals that the model current, when accounting for both gear backlash and nonlinear friction, most closely approximates the actual current. Table III presents the mean AE (MAE) and root-mean-square error (RMSE) between the model's current and the actual current for the four scenarios. As shown in the table, Case 1 has the smallest deviation from the actual current, followed by Case 2 and Case 3, while the linear model shows the largest error.

Therefore, the experimental results indicate that models incorporating nonlinear factors exhibit higher accuracy. Additionally, nonlinear friction has a more significant impact on the model's accuracy, while the influence of gear backlash is relatively smaller. This phenomenon is partly attributed to the relatively small gear backlash in the switch machine used for data collection, which limited its impact on the model. This ablation experiment confirms that the model established in this article demonstrates excellent accuracy and further highlights the contributions of various nonlinear factors to the model's precision, providing a basis for continued model optimization.

B. State Estimation Precision Validation Experiment

In this section, state estimation simulation experiments are carried out to validate the performance of the proposed improved EKF based on statistical information under the third-order Taylor expansion. The simulation experiment takes the nonlinear model of the RTS established in this article as the object, divided into two cases of normal and abnormal working conditions. Furthermore, it is assumed that the system noise and the measurement noise are a set of mutually independent zero-mean Gaussian white noise sequences, and the initial state is set to zero.

Denote the improved EKF method proposed in this article as 3oEKF, which includes both second-order and third-order statistical features, and the improved EKF that only contains second-order statistical features as 2oEKF. These two methods are compared with existing filtering methods for state estimation of nonlinear systems, such as EKF, UKF, AEKF, and AUKF. The simulation results are based on 1000 Monte Carlo statistics. The simulation experiments use the MAE, RMSE, and mse as the evaluation metrics for each filter.

1) Normal Working Condition: To maintain consistency with the sampling rate of the real current data used for model validation in this article, the sampling time interval is set to 0.04 s during the normal operation of the turnout system, with the turnout operation being fixed control. Taking the two critical variables involved in the switching machine, i.e., current and angular velocity, as the estimation objects, the actual values obtained based on the model and the estimates obtained from each filtering method are shown in Figs. 5 and 6. The operation time contains 94 sampling points, which is 3.76 s. The RMSE and enhancement of 2oEKF, 3oEKF, and other methods are shown in Tables IV and V. By taking each data point as an object, the MAE and mse values are calculated. The resulting MAE and mse variation curves for each filtering method are shown in Figs. 7 and 8.

As seen from Table V and Fig. 5, the performance of the 2oEKF, which utilizes only second-order statistical

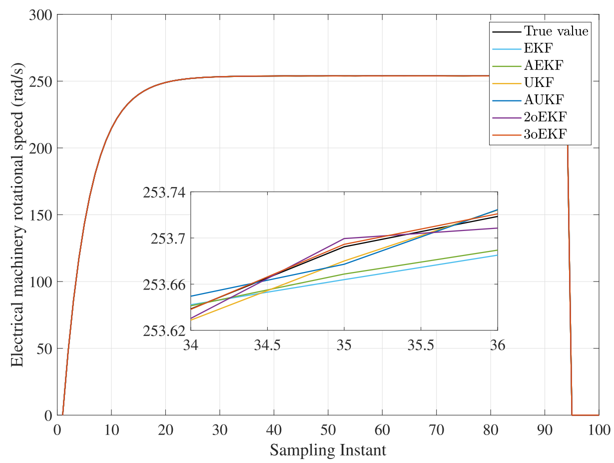


Fig. 6. Angular velocity state estimation under normal operating conditions.

TABLE IV
RMSE Comparison of Current Value Filtering Results

Filter type	RMSE	3oEKF improvement rate to
EKF	0.0117	33.33%
AEKF	0.0108	27.78%
UKF	0.0102	23.53%
AUKF	0.0101	22.77%
2oEKF	0.0093	16.13%
3oEKF	0.0078	/

TABLE V
RMSE Comparison of Angular Velocity Filtering Results

Filter type	RMSE	3oEKF improvement rate to
EKF	0.0284	38.73%
AEKF	0.0254	31.50%
UKF	0.0269	35.32%
AUKF	0.0217	19.82%
2oEKF	0.0196	11.22%
3oEKF	0.0174	/

information, has surpassed that of the EKF and UKF in state estimation of currents. The RMSE of the 2oEKF is reduced by 20.51% compared to the basic EKF. The performance of 3oEKF with the addition of third-order statistical information is even improved by 33.33% compared to EKF. It significantly outperforms other methods in terms of filtering performance. The same conclusion can be obtained by analyzing the MAE and mse curves in Fig. 7.

From Fig. 6, it can be seen that the angular velocity of the motor in the RTS model established in this article is stabilized at about 253.9 rad/s, which is 2425.8 r/min. The rotational speed value is in line with the motor's calibrated rotation speed of the rutting machine of this model (≥2400 r/min). The performance analysis in Table V and Fig. 8 shows that in the state estimation of angular velocity, the filtering accuracies of 2oEKF and 3oEKF are also more prominent than the other methods, which show a strong filtering performance. This enhancement is mainly due to their accuracy in dealing with nonlinear systems.

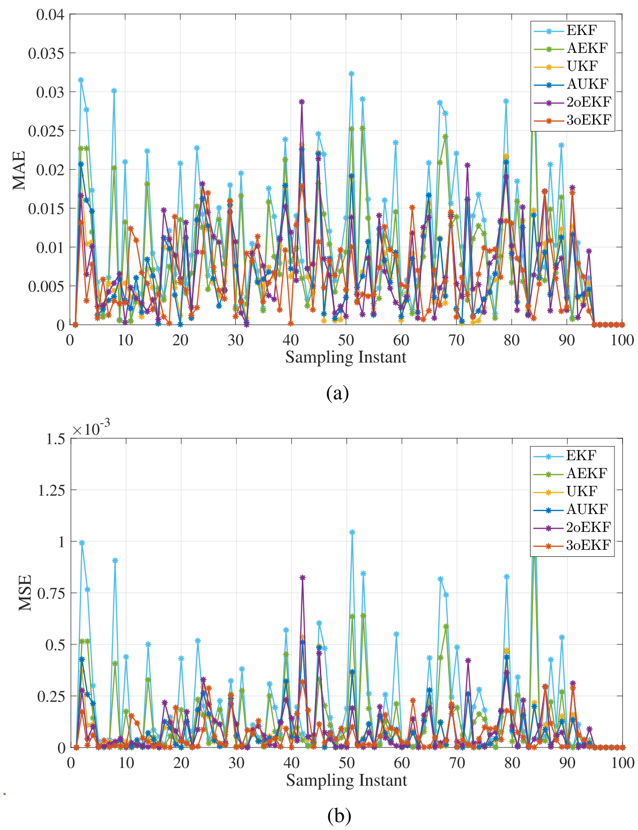


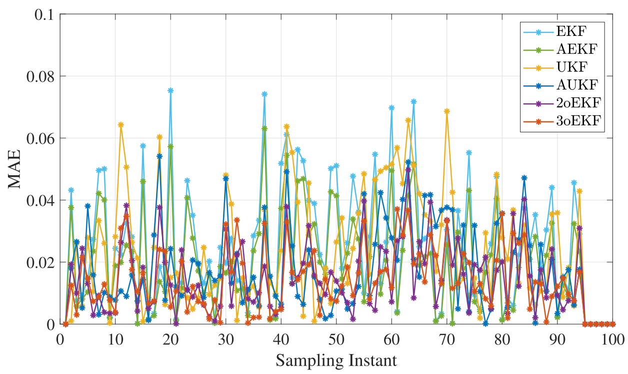
Fig. 7. Current state estimation evaluation. (a) MAE of estimated values at each point. (b) MSE of estimated values at each point.

TABLE VI
RMSE Comparison of Current Value Filtering Results Under Abnormal Operating Conditions

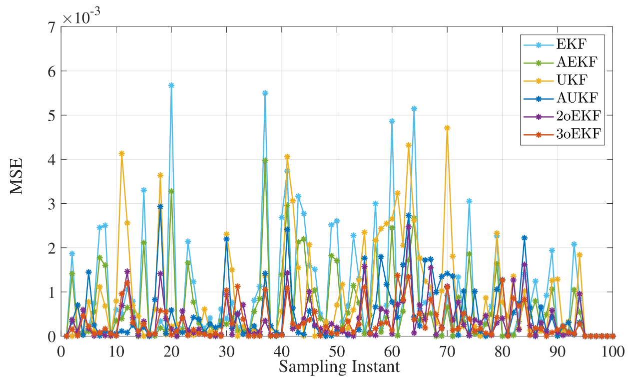
Filter type	RMSE	3oEKF improvement rate to
EKF	0.0092	28.26%
AEKF	0.0084	21.43%
UKF	0.0074	10.81%
AUKF	0.0075	12.00%
2oEKF	0.0078	15.38%
3oEKF	0.0066	/

2) Abnormal Working Condition: Abnormal working conditions of the RTS can lead to changes in its operating status. In this experiment, the viscous friction coefficient of the mechanical part of the motor is set to change abruptly from 0.03 to 0.05 at the 76th sampling point, and the obtained state estimation results are shown in Figs. 9 and 10. It can be seen from the figure that with the sudden change in A, the current jumped from near 0.3 to 0.9 A, and the angular velocity decreased from near 253.5 to near 199.5 rad/s. This change also caused the switch machine operation to continue until the 141st sampling point, i.e., 5.64 s, which exceeds the calibrated changeover time (less than or equal to 3.8 s) for this model of switch machine.

Tables VI and VII give the RMSE of each filter under abnormal working conditions. In the current state estimation, the 2oEKF method is close to the performance of UKF and AUKF, and 3oEKF is still the best result, with an



(a)



(b)

Fig. 8. Angular velocity state estimation evaluation. (a) MAE of estimated values at each point. (b) MSE of estimated values at each point.

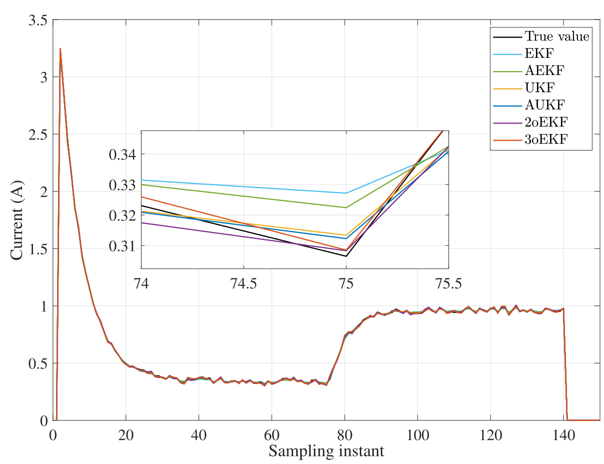


Fig. 9. Current state estimation under abnormal operating conditions.

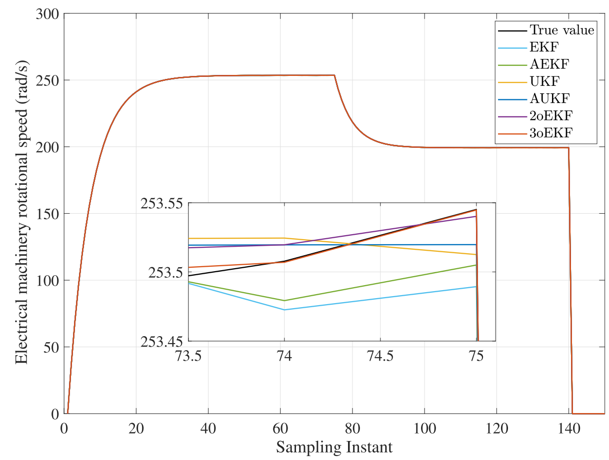
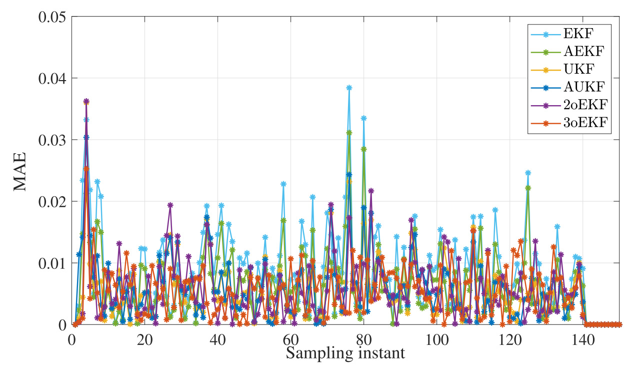
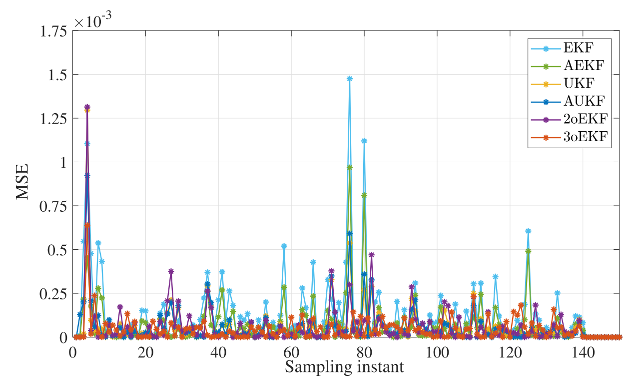


Fig. 10. Angular velocity state estimation under abnormal operating conditions.



(a)



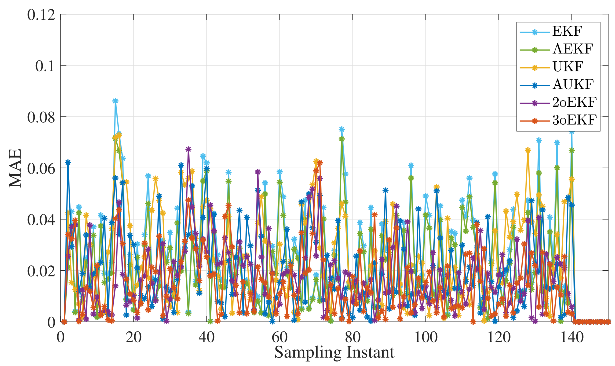
(b)

Fig. 11. Current state estimation evaluation. (a) MAE of estimated values at each point. (b) MSE of estimated values at each point.

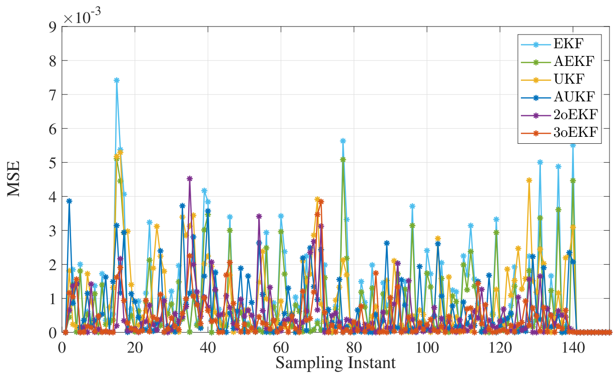
improvement of 28.26% compared to EKF. In angular velocity state estimation, 2oEKF and 3oEKF outperform the other filters, improving the RMSE compared to the original EKF by 30.50% and 38.05%, respectively.

Figs. 11 and 12 show the MAE and mse curves of the current and angular velocity estimation results under abnormal operating conditions, respectively. It can be observed that both 2oEKF and 3oEKF exhibit lower MAE and mse values at most sampling points. Notably, the estimation accuracy of 3oEKF is higher compared to other filtering methods at several sampling points following the occurrence of the mutation.

The comparison between normal and abnormal working conditions demonstrates that the 3oEKF method proposed in this article exhibits superior performance in the state estimation of the RTS model. This approach leverages the second-order and third-order statistical information from the Taylor expansion to enhance the filtering performance of the EKF. The experimental results indicate an improvement of over 28%. Hence, the method proposed in this



(a)



(b)

Fig. 12. Angular velocity state estimation evaluation. (a) MAE of estimated values at each point. (b) MSE of estimated values for each point.

TABLE VII
RMSE Comparison of Angular Velocity Value Filtering Results Under Abnormal Operating Conditions

Filter type	RMSE	3oEKF improvement rate to
EKF	0.0318	38.05%
AEKF	0.0281	29.89%
UKF	0.0305	35.41%
AUKF	0.0259	23.94%
2oEKF	0.0221	10.86%
3oEKF	0.0197	/

article is highly suitable for high-precision state estimation of RTS.

V. Conclusion

In this article, a nonlinear model for the RTS is presented in the first-ever establishment using the BG modeling method. The model represents an RTS driven by a ZD6-A dc switch machine, considering the effects of nonlinear friction and gear backlash. The accuracy of the constructed model is verified using BG simulation software with actual current data and nominal values of the switch machine. Additionally, to improve state estimation accuracy for the nonlinear model, second- and third-order statistical information from the Taylor expansion of the nonlinear function is incorporated into the filter design using the ULS method. This approach significantly improves filter performance. Finally, the superiority of the proposed filtering method over other filtering methods is demonstrated by analyzing the state estimation results of current and angular velocity under normal and abnormal operating conditions of the RTS.

While this study focuses on nonlinear modeling and filtering methods for RTSs, challenges remain in capturing complex nonlinearities, adapting to diverse conditions, and addressing scenarios with measurement nonlinearities and non-Gaussian noise. Future research will extend the proposed methods to other types of turnout systems and explore the application of physics-informed machine learning to enhance model physical consistency and data utilization efficiency [28], [29]. By integrating physical dynamics and sensor data, these advancements aim to further improve model accuracy and robustness, providing robust support for fault diagnosis, system optimization, and safety analysis, thereby contributing to the intelligent development of RTS.

Appendix I
Performance Analysis of Higher Order Terms

The state nonlinear equation represented by (18) can be expanded into an r-order Taylor series, yielding

$$\boldsymbol{x}_{\{r\}}(k+1) = \boldsymbol{\phi}(\hat{\boldsymbol{x}}^{(1)}(k\,|k)) + \sum_{i=1}^{r} \boldsymbol{A}^{(i)}(k)\tilde{\boldsymbol{x}}^{(i)}(k\,|k) + \boldsymbol{B}(k)\boldsymbol{u}(k) + \boldsymbol{w}(k). \tag{40}$$

Its one-step predicted estimate is

$$\hat{\boldsymbol{x}}_{\{r\}}(k+1\,|k) = \boldsymbol{\phi}(\hat{\boldsymbol{x}}^{(1)}(k\,|k)) + \sum_{i=1}^{r} \boldsymbol{A}^{(i)}(k)\hat{\tilde{\boldsymbol{x}}}^{(i)}(k\,|k) + \boldsymbol{B}(k)\boldsymbol{u}(k). \tag{41}$$

Thus, its estimation error should be

$$\begin{aligned}\tilde{\boldsymbol{x}}_{\{r\}}(k+1\,|k) &= \boldsymbol{x}_{\{r\}}(k+1) - \hat{\boldsymbol{x}}_{\{r\}}(k+1\,|k)\\ &= \sum_{i=1}^{r} \boldsymbol{A}^{(i)}(k)[\tilde{\boldsymbol{x}}^{(i)}(k\,|k) - \hat{\tilde{\boldsymbol{x}}}^{(i)}(k\,|k)] + \boldsymbol{w}(k)\\ &= \sum_{i=1}^{r} \boldsymbol{A}^{(i)}(k)\tilde{\tilde{\boldsymbol{x}}}^{(i)}(k\,|k) + \boldsymbol{w}(k)\end{aligned} \tag{42}$$

where $\tilde{\tilde{\boldsymbol{x}}}^{(i)}(k|k) = \tilde{\boldsymbol{x}}^{(i)}(k|k) - \hat{\tilde{\boldsymbol{x}}}^{(i)}(k|k)$. Further, the covariance matrix of $\tilde{\boldsymbol{x}}_{\{r\}}(k+1|k)$ is calculated as follows:

$$\begin{aligned}\boldsymbol{P}_{\{r\}}(k+1\,|k) &= E\left\{\tilde{\boldsymbol{x}}_{\{r\}}(k+1\,|k)\tilde{\boldsymbol{x}}_{\{r\}}(k+1\,|k)^{\mathrm{T}}\right\}\\ &= E\left\{\left[\sum_{i=1}^{r} \boldsymbol{A}^{(i)}(k)\tilde{\tilde{\boldsymbol{x}}}^{(i)}(k\,|k) + \boldsymbol{w}(k)\right]\right.\\ &\qquad \left.\cdot\left[\sum_{i=1}^{r} \boldsymbol{A}^{(i)}(k)\tilde{\tilde{\boldsymbol{x}}}^{(i)}(k\,|k) + \boldsymbol{w}(k)\right]^{\mathrm{T}}\right\}\\ &= \sum_{i=1}^{r} \boldsymbol{A}^{(i)}(k)\boldsymbol{P}_{\tilde{\tilde{x}}\tilde{\tilde{x}}}{}^{(i,i)}(k\,|k)\boldsymbol{A}^{(i)}(k)^{\mathrm{T}}\\ &\quad + \boldsymbol{Q}(k).\end{aligned} \tag{43}$$

Similarly, based on the nonlinear state model, the l-order Taylor expansion ($l < r$) is expressed as

$$\begin{aligned}\tilde{\boldsymbol{x}}_{\{l\}}(k+1\,|k) &= \boldsymbol{x}_{\{l\}}(k+1) - \hat{\boldsymbol{x}}_{\{l\}}(k+1\,|k)\\ &= \sum_{i=1}^{l} \boldsymbol{A}^{(i)}(k)\tilde{\tilde{\boldsymbol{x}}}^{(i)}(k\,|k) + \boldsymbol{w}(k).\end{aligned} \tag{44}$$

The one-step measurement prediction estimation error, referenced to the r-order Taylor expansion, can be expressed as

$$\begin{aligned}\tilde{\boldsymbol{x}}_{\{l\}}(k+1\,|k) &= \boldsymbol{x}_{\{r\}}(k+1) - \hat{\boldsymbol{x}}_{\{l\}}(k+1\,|k)\\ &= \sum_{i=1}^{l} \boldsymbol{A}^{(i)}(k)\left[\tilde{\boldsymbol{x}}^{(i)}(k\,|k) - \hat{\tilde{\boldsymbol{x}}}^{(i)}(k\,|k)\right]\\ &\quad + \sum_{i=l+1}^{r} \boldsymbol{A}^{(i)}(k)\tilde{\boldsymbol{x}}^{(i)}(k\,|k) + \boldsymbol{w}(k)\\ &= \sum_{i=1}^{r} \boldsymbol{A}^{(i)}(k)\tilde{\tilde{\boldsymbol{x}}}^{(i)}(k\,|k)\\ &\quad + \sum_{i=l+1}^{r} \boldsymbol{A}^{(i)}(k)\tilde{\boldsymbol{x}}^{(i)}(k\,|k) + \boldsymbol{w}(k).\end{aligned} \tag{45}$$

Thus, the covariance matrix of the prediction error $\tilde{\boldsymbol{x}}_{\{l\}}(k+1|k)$ is

$$\begin{aligned}\boldsymbol{P}_{\{l\}}(k+1\,|k) &= E\{\tilde{\boldsymbol{x}}_{\{l\}}(k+1\,|k)\tilde{\boldsymbol{x}}_{\{l\}}(k+1\,|k)^{\mathrm{T}}\}\\ &= \sum_{i=1}^{r} \boldsymbol{A}^{(i)}(k)\boldsymbol{P}_{\tilde{\tilde{\boldsymbol{x}}}\tilde{\tilde{\boldsymbol{x}}}^{(i,i)}}(k\,|k)\\ &\quad + \sum_{i=l+1}^{r} \boldsymbol{A}^{(i)}(k)\hat{\tilde{\boldsymbol{x}}}^{(i)}(k\,|k)\hat{\tilde{\boldsymbol{x}}}^{(i)}(k\,|k)^{\mathrm{T}}\boldsymbol{A}^{(i)}(k)^{\mathrm{T}}\\ &\quad + \boldsymbol{Q}(k).\end{aligned} \tag{46}$$

Comparing (43) and (46), if $r = l + 1$, then

$$\begin{aligned}\boldsymbol{P}_{\{l\}}(k+1\,|k) &= \boldsymbol{P}_{\{l+1\}}(k+1\,|k)\\ &\quad + \boldsymbol{A}^{(l+1)}(k)\hat{\tilde{\boldsymbol{x}}}^{(l+1)}(k\,|k)\hat{\tilde{\boldsymbol{x}}}^{(l+1)}(k\,|k)^{\mathrm{T}}\\ &\quad \times \boldsymbol{A}^{(l+1)}(k)^{\mathrm{T}}.\end{aligned} \tag{47}$$

Since $\boldsymbol{A}^{(l+1)}(k)\hat{\tilde{\boldsymbol{x}}}^{(l+1)}(k|k)\hat{\tilde{\boldsymbol{x}}}^{(l+1)}(k|k)^{\mathrm{T}}\boldsymbol{A}^{(l+1)}(k)^{\mathrm{T}} \geq 0$, it follows that:

$$P_{\{l\}}(k+1\,|k) \geq P_{\{l+1\}}(k+1\,|k), \quad l = 1, 2, \ldots, r-1. \tag{48}$$

The covariance matrix of the state estimation error can be expressed as

$$\begin{aligned}\boldsymbol{P}_{\{l\}}(k+1\,|k+1)^{-1} &= \boldsymbol{P}_{\{l\}}(k+1\,|k)^{-1}\\ &\quad + \boldsymbol{H}(k+1)^{\mathrm{T}}\boldsymbol{R}(k+1)^{-1}\boldsymbol{H}(k+1)\\ &\leq \boldsymbol{P}_{\{l+1\}}(k+1\,|k)^{-1}\\ &\quad + \boldsymbol{H}(k+1)^{\mathrm{T}}\boldsymbol{R}(k+1)^{-1}\boldsymbol{H}(k+1)\\ &= \boldsymbol{P}_{\{l+1\}}(k+1\,|k+1)^{-1}\end{aligned} \tag{49}$$

TABLE VIII
COMPLEXITY OF SIX FILTERING ALGORITHMS

Methods	Time complexity	Space complexity
EKF	$O(n^3)$	$O(n^2)$
AEKF	$O(n^3)$	$O(n^2)$
UKF	$O(n^3)$	$O(n^2)$
AUKF	$O(n^3)$	$O(n^2)$
2oEKF	$O(n^4)$	$O(n^2)$
3oEKF	$O(n^5)$	$O(n^3)$

then

$$P_{\{l\}}(k+1\,|k+1) \geq P_{\{l+1\}}(k+1\,|k+1) \quad l = 1, 2, \ldots, r-1. \tag{50}$$

As a result, according to (50), as higher order terms of the nonlinear state model are progressively utilized, the error covariance of the designed filter gradually decreases, leading to higher state estimation accuracy and improved filter performance.

APPENDIX II
ALGORITHM COMPLEXITY ANALYSIS

In terms of computational cost, the complexity of an algorithm is typically evaluated through two metrics: time complexity and space complexity. Time complexity refers to the amount of time required during the execution of an algorithm, expressed as a function of the problem size, which describes the growth trend of the number of basic operations needed for execution. Space complexity refers to the amount of memory required during the execution of an algorithm, also expressed as a function of the problem size, describing the growth trend of additional memory usage during execution. Based on computational analysis, the complexity statistics of the six filtering algorithms—EKF, AEKF, UKF, AUKF, 2oEKF, and 3oEKF—are presented in Table VIII.

In the context of filtering algorithms, the complexity term n can be interpreted as the dimensionality of the state space. As shown in Table VIII, the computational complexity of EKF, AEKF, UKF, and AUKF is comparable, while the time complexity of 2oEKF and 3oEKF increases progressively. The space complexity of 3oEKF also rises. This increase is primarily due to the inclusion of second-order and third-order Taylor expansion computations in 2oEKF and 3oEKF, along with the storage of additional higher order derivative terms, contributing to the growth in computational complexity.

Additionally, Table IX records the average code execution time over 500 runs for the six filtering methods. The code was executed in MATLAB R2023a on a computer equipped with an Intel Core i5-8265U CPU at 1.8 GHz and 8 GB of RAM. The table shows that while the computational complexity of UKF and AUKF is not significantly different from that of EKF and AEKF, their code execution times are noticeably longer. The primary reason is that the UKF requires the generation and processing of a greater number of sigma points for each state dimension, resulting in significantly more

TABLE IX
AVERAGE CODE EXECUTION TIME OF SIX FILTERING ALGORITHMS

Methods	Average time (s)
EKF	0.00081
AEKF	0.00101
UKF	0.00602
AUKF	0.00608
2oEKF	0.00494
3oEKF	0.00619

evaluations of nonlinear functions, matrix operations, and covariance reconstruction, which leads to considerably longer execution times compared to the EKF. At the same time, although the code execution time of the 3oEKF is slightly longer than that of the EKF, it is relatively close to that of the UKF.

According to the results, while the computational complexity of the 3oEKF is higher than that of the EKF and UKF, its code execution time remains comparable to that of the UKF.

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